

Deep Networks

MLP/FNN to Universal Function Approximation

Introduction to Deep Learning · Class 9

10 August

Outline

The Multilayer Perceptron, Formally

Universal Approximation, Revisited

How Wide Must a Network Be?

What This Settles

The Multilayer Perceptron, Formally

Notation used throughout this course

$\mathbf{x} \in \mathbb{R}^D$	input / features	$\phi_j(\mathbf{x})$	fixed basis function
$y, \hat{y}$	target, model prediction	$\mathcal{K} \subset \mathbb{R}^D$	a compact set
$\mathbf{z}^{(l)} = h(\mathbf{a}^{(l)})$	units of hidden layer l	$C(\mathcal{K})$	continuous functions on $\mathcal{K}$
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)}$	weights and biases	$\ f\ _\infty$	$\sup_{\mathbf{x} \in \mathcal{K}} f(\mathbf{x}) $
L, M	depth, width	ε	target accuracy
W	total number of parameters	$\mathcal{N}_{L,M}$	functions a network can compute

Symbols follow **Bishop & Bishop (2024)**. W counts every weight and bias; $\mathcal{N}_{L,M}$ is the set of functions a network of that shape can compute.

Where Class 7 left us

- Composition **multiplies** linear pieces; width only **adds** them
- Universal approximation holds in D dimensions for any non-polynomial activation, via ridge functions
- ReLU networks compute **exactly** the continuous piecewise-linear functions — shallow or deep
- The sawtooth needs $O(L)$ units deep and $2^L - 1$ shallow

All of that was about **what** a network can represent. None of it says **how much network** it takes.

Three questions, and where they are answered

	Question	Answered by	
1	Can one hidden layer reach any function at all?	Cybenko, Hornik, Leshno	§2
2	How fast does the error fall as the network grows?	Barron; DeVore et al.	§2
3	How wide must a network be before it works at all?	Lu et al.; Hanin & Sellke	§3

A fourth question — how the budget should be *split* between depth and width — is the whole of **Class 10**.

The multilayer perceptron

Before we can count anything, we need to fix exactly what we are counting.

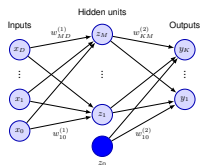


Figure from Bishop & Bishop (2024), Fig. 6.9.

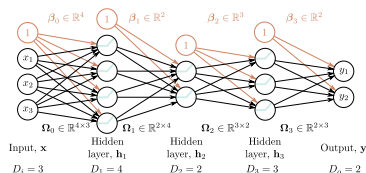


Figure from Prince (2023), Fig. 4.6: the same network with L hidden layers and their dimensions.

$$\mathbf{z}^{(0)} = \mathbf{x}, \quad \mathbf{z}^{(l)} = h(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}), \quad \hat{\mathbf{y}} = \mathbf{W}^{(L+1)}\mathbf{z}^{(L)} + \mathbf{b}^{(L+1)}$$

$\mathcal{N}_{L,M}$ denotes the functions this computes, with $W = M(D+1) + (L-1)M(M+1) + (M+1)$ parameters.

Why not just use fixed basis functions?

Before calling learned features progress, see what goes wrong without them.

Tile the input space, one basis function per cell (B&B, 2024, §6.1):

$$\hat{y}(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x}), \quad \phi_j \text{ centred on cell } j$$

Proposition 1 (The grid blows up)

Partition each of the D axes into S intervals. The tensor-product grid then has S^D cells, so the model has S^D parameters — exponential in the input dimension.

Proof. A cell is one interval chosen per axis, so there are $|\{1, \dots, S\}^D| = S^D$. □

With $S = 10$ and a 32×32 image ($D = 1024$): 10^{1024} cells.

The curse of dimensionality, drawn

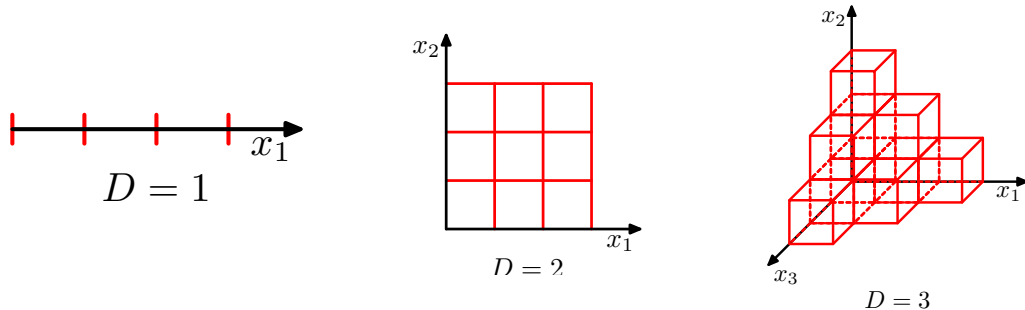
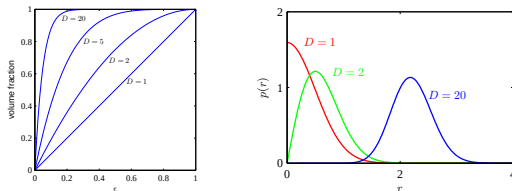


Figure from Bishop & Bishop (2024), Fig. 6.3: a regular grid over the input space in $D = 1, 2$ and 3 . The number of cells grows as S^D , so each extra input dimension multiplies the number of parameters.

Every cell needs its own parameter — and its own data to fit it.

Two more faces of the same problem

High-dimensional space is strange in two further ways that matter later.



Figures from Bishop & Bishop (2024), Figs. 6.4 and 6.5. Left: the volume of a sphere lying in an outer shell of thickness ϵ . Right: a Gaussian's radial density, whose mass migrates to a shell at radius $\approx \sqrt{D}$.

SHALLOW fixed ϕ_j : one parameter per cell, S^D of them

DEEP learned features: units *move* to where the data is

Universal Approximation, Revisited

What “universal” means

Networks are supposed to escape this. First, say precisely what that claim asserts.

Fix a compact $\mathcal{K} \subset \mathbb{R}^D$ and use the supremum norm $\|f\|_\infty = \sup_{\mathbf{x} \in \mathcal{K}} |f(\mathbf{x})|$.

Definition

A family $\mathcal{F} \subset C(\mathcal{K})$ is **universal** (dense) if for every $f \in C(\mathcal{K})$ and every $\varepsilon > 0$ there is $g \in \mathcal{F}$ with $\|f - g\|_\infty < \varepsilon$.

- Note what is quantified: $\mathcal{F}$ is the family with **unbounded** width, so “ g exists” may mean a network of astronomical size
- Density says nothing about **how fast** the error falls as the network grows — that needs a **rate**

The classical statement

With the definition in place, the theorem is short.

Theorem 2 (Universal approximation; Cybenko 1989, Hornik 1991, Leshno et al. 1993)

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. The family

$$\mathcal{N}_{1,\infty} = \left\{ \mathbf{x} \mapsto \sum_{j=1}^M w_j^{(2)} h(\mathbf{w}_j^{(1)\top} \mathbf{x} + w_{j0}^{(1)}) + w_0^{(2)} : M \in \mathbb{N} \right\}$$

is dense in $C(K)$ for every compact K if and only if h is not a polynomial.

- Proved in Class 7 by reduction to one dimension through **ridge functions**
- **One hidden layer is enough** — so universality alone can never be an argument for depth

Universal approximation, illustrated

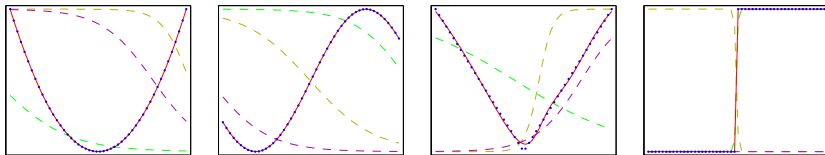


Figure from Bishop & Bishop (2024), Fig. 6.10: one two-layer network with three $\tanh$ hidden units, fitted to x^2 , $\sin(2\pi x)$, $|x|$ and the step $\mathbb{I}[x > 0]$. Dashed are the individual hidden-unit contributions; solid is the network output.

- Even three units get close; the theorem says the error goes to zero as the width grows
- The step function shows the limit: a continuous network can only approximate a discontinuity, never match it

Two objects the next two theorems need

Density says the error reaches zero. To ask how fast, we must first say which functions we mean.

The Barron norm

$$C_f = \int \|\omega\| |\tilde{f}(\omega)| d\omega$$

$\tilde{f}(\omega)$ is the Fourier transform: how much of f sits at frequency ω . Weighting by $\|\omega\|$ **charges a function** for its high-frequency content.

A smoothness ball

$$\mathcal{F} = \{f: \|\partial^\alpha f\|_\infty \leq 1 \ \forall |\alpha| \leq s\}$$

Every derivative up to order s is bounded by one.
This is the Sobolev ball $W^{s,\infty}$: **all functions of a** given smoothness, taken together.

One measures a **single** function's wiggleness, the other a **whole class** — which is why the two bounds point opposite ways. A bound over a ball is worst-case.

A rate that does not see the dimension

Take a function with small C_f . How quickly does a one-hidden-layer network close in on it?

Theorem 3 (Barron, 1993)

Let $f: \mathcal{K} \rightarrow \mathbb{R}$ have finite Barron norm $C_f < \infty$. Then for every M there is a one-hidden-layer sigmoidal network $\hat{y}_M$ with M units and

$$\int_{\mathcal{K}} (f(\mathbf{x}) - \hat{y}_M(\mathbf{x}))^2 \mu(d\mathbf{x}) \leq \frac{(2C_f r)^2}{M},$$

with r the radius of $\mathcal{K}$. The bound is *independent of D* .

- **Proof sketch.** Write f as a continuous mixture of ridge functions, with total mass controlled by C_f
- So f is in the closed convex hull of scaled ridge functions of norm $\leq 2C_f r$
- Maurey's lemma: a point of the convex hull of norm- b elements lies within $b/\sqrt{M}$ of an average of M of them
- That average is an M -unit network

□

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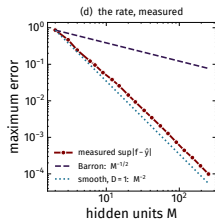
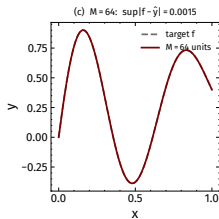
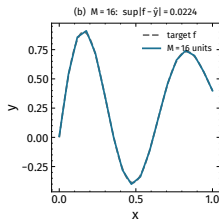
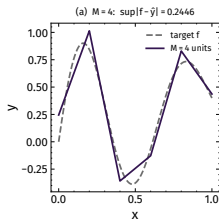
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- **That average is an M -unit network**

□

The rate, measured



- (a-c) the best M -unit fit to one smooth target as M grows
- (d) the measured error against two references
- Barron's $M^{-1/2}$ is a **worst-case guarantee** over a whole class; a single smooth target in $D = 1$ does much better, here M^{-2}
- A theorem gives a ceiling on the error, not a prediction of it

And a lower bound that does

Barron's class is generous to us. What happens if we ask for a guarantee over all functions of a given smoothness instead?

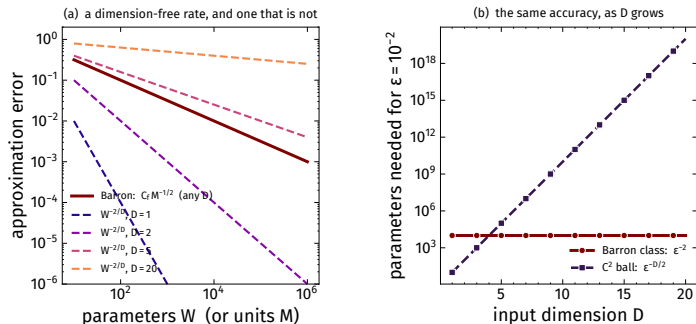
Theorem 4 (DeVore, Howard & Micchelli, 1989)

*Let $\mathcal{F}$ be the unit ball of the Sobolev space $W^{s,\infty}([0,1]^D)$. If an approximation scheme uses W parameters chosen **continuously** from the target, then its worst-case error over $\mathcal{F}$ is at least $cW^{-s/D}$.*

- Smoothness s buys accuracy; dimension D takes it away. To halve the error in D dimensions you need $2^{D/s}$ times the parameters
- It limits **every** scheme with continuous parameter selection, not networks in particular
- No contradiction with Barron: his class is **much smaller** than a Sobolev ball

The curse is real. It is escaped only by targets with extra structure — which is exactly what real data has.

The two bounds side by side



(a) the dimension-free Barron rate against the Sobolev lower bound for several D : for $D = 1$ or 2 smoothness wins, but by $D = 20$ the Barron bound is far better. (b) parameters needed for $\epsilon = 10^{-2}$: constant in the Barron class, exponential in D on a C^2 ball.

How Wide Must a
Network Be?

Width as a threshold: the upper bound

We know the cost of accuracy. Now: how to spend the budget. Width behaves quite unlike depth.

Theorem 5 (Width-bounded universality; Lu, Pu, Wang, Hu & Wang, 2017)

*ReLU networks of width $D + 4$ and **arbitrary depth** are dense in $L^1(\mathbb{R}^D)$: for every Lebesgue-integrable f and every $\varepsilon > 0$ there is such a network with $\int |f - \hat{y}| \, d\mathbf{x} < \varepsilon$.*

- Four units more than the input dimension is **all** the width that is ever needed, provided the network may be deep
- Note the norm: density in L^1 , not uniform approximation

Width as a threshold: the lower bound

Four units of slack sounds arbitrary. It is not — just below it, everything fails.

Theorem 6 (Width lower bound; Hanin & Sellke 2017, Johnson 2019)

ReLU networks of width $\leq D$ are not universal, however deep. There are continuous functions they cannot approximate to better than a fixed constant.

Why it holds. A ReLU network of width $\leq D$ never expands the space it maps through, so all of its **level sets are unbounded**. A function with a bounded level set — say $\|x\|$, whose level sets are spheres — cannot be approximated.

Width is a threshold: below D nothing works, just above it everything does. **Depth is a resource.**

What This Settles

What each theorem actually gives

Ten results in one lecture. What each one is really claiming:

Result	Gives	Does not give
Cybenko / Hornik / Leshno	density with one hidden layer, any non-polynomial h	any bound on the width
Barron (1993)	$O(C_f/\sqrt{M})$, free of D	anything outside the Barron class; C_f may grow with D
DeVore et al. (1989)	$W^{-s/D}$ is unbeatable on a Sobolev ball	a statement about one particular target

What none of them say

And the gap between all this and a working network:

- **Nothing about optimisation.** Every result is about the *existence* of weights; finding them is Classes 11–14
- **Nothing about generalisation.** The capacity bounds are worst-case over all data distributions
- **Architecture matters more than the counts suggest.** Convolution and attention change the function class itself

SHALLOW universal, with a vacuous width bound

DEEP the useful question is what the budget *buys*
— Class 10

Takeaways

1. Fixed basis functions need S^D parameters. Networks avoid this by **placing their features**, not tiling the space
2. One hidden layer is already universal, for any non-polynomial activation — so **universality alone never argues for depth**
3. Rates are what matter. **Barron**: $O(C_f/\sqrt{M})$, free of dimension, on a restricted class. **DeVore et al.**: $W^{-s/D}$ is unbeatable on a smoothness ball
4. **Width is a threshold**: $\leq D$ never works however deep, and $D + 4$ always does given enough depth — so width is something to clear, not something to tune

Class 10: universality and rates say nothing about *how the budget is shaped*. Next: depth against width.

References

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- S. J. D. Prince. *Understanding Deep Learning*. MIT Press, 2023. Ch. 4 — the universal approximation theorem and its width version. The depth- and width-efficiency results from the same notes are Class 10.
- G. Cybenko (1989); K. Hornik (1991); M. Leshno, V. Lin, A. Pinkus & S. Schocken, *Neural Networks* 6:861 (1993).
- A. R. Barron, *IEEE Trans. Inf. Theory* 39:930 (1993) — dimension-free approximation rates.
- R. DeVore, R. Howard & C. Micchelli, *Numer. Math.* 63:469 (1989) — nonlinear widths.
- Z. Lu, H. Pu, F. Wang, Z. Hu & L. Wang, *NeurIPS* (2017); B. Hanin & M. Sellke (2017); J. Johnson, *ICLR* (2019) — width bounds.
- Figures 6.2–6.5, 6.9 and 6.10 reproduced from Bishop & Bishop (2024); Figures 4.6 and 4.9 from Prince (2023), <https://udlbook.github.io/udlbook/>, CC BY-NC-ND. The rate and capacity plots have no textbook counterpart and are generated by `../B_Recreated_Figures/src/build_all.py`.

Questions?